

Data Science and Data Analytics

Getting started

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Course design

Welcome to Data Science and Data Analytics!

- This is **B-GV-12 Data Science and Data Analytics**.
- The goal of this course is to teach you...
 - about the nature and role of data science in a data-driven world.
 - about the data science workflow.
 - to use open-source software to analyse, visualize and model data from various sources.
 - about different machine learning algorithms.
 - to implement your own data science projects.
 - to communicate the results of your analyses effectively.
 - and much more...

Lectures

- With a few exceptions, lectures will be held on **Fridays from 13:30 to 16:30**.
- Please regularly check your course schedule to not miss any lectures.
- Lectures will consist of theory and practice discussed with the help of **slides** as well as **live coding sessions**.
- You are highly encouraged to **actively participate!**
- Exercises to practice what you have learned will be provided (but not graded).

Grading

- There will be **no** final exam in this course!
- Instead, grading will be based on **group projects**:
 - Teams of 4 - 5 students will initiate and design a small data science project autonomously.
 - Each group will:
 - identify a business case mimicking a real-world research problem and an accompanying available data set.
 - formulate research questions on the basis of the chosen data set.
 - perform analyses using the concepts and methods learned throughout this course.

Grading

- Grading will therefore be based on the following components:
 - **Group project report** (5-10 pages per group member): **65 %**
 - **Group presentation** (≤ 20 mins/group and ≥ 2 mins/group member): **25 %**
 - **Peer review**: **10 %**
- In line with the usual grading scheme, grades will be given as follows:

Percentage	Grade
95 - 100 %	1,0
90 - 94 %	1,3
85 - 89 %	1,7
80 - 84 %	2,0
75 - 79 %	2,3
70 - 74 %	2,7

Percentage	Grade
65 - 69 %	3,0
60 - 64 %	3,3
55 - 59 %	3,7
50 - 54 %	4,0
below 50 %	5,0

Grading: Group project

- The **structure** of the group project (reflected in report and presentation) should be something like this:
 - Introduction / Motivation and research question
 - Data (sources, description, statistics, visualizations, ...)
 - Models and model evaluation
 - Results
 - Discussion and comments
- Aspects that will influence your grade will be: the originality of the question, understanding of the business case, data and methods, correctness of application, thoroughness of evaluation, creativity and quality of report and presentation (both verbal and visual)
- Deductions will be made for purely AI-generated contributions.

Grading: Group project

Choice of topic: while you are completely free in your choice of topic in the group, here are some areas of suggestion:

- Finance / Economics / Marketing
- Text analysis
- Entertainment (in particular: movies and music)
- Social network analysis
- Social sciences

Sources for data sets: while you are again free also in your choice of data set, good places to get you started are:

- [Statistik Austria](#)
- [Kaggle](#)
- [UCI Machine learning repository](#)
- [World bank](#)
- [EU](#)
- ...

Caution

When selecting a data set, make sure, you are **allowed** to use this data for the purposes of your project! When selecting from the sources given, this should generally be ensured.

Grading: Peer review

- After final project presentations, each **individual** student will be asked to write a **peer review** of one of the other groups' projects.
- Each student will be randomly assigned **two projects**, out of which **one** should be reviewed (based on their presentation only).
- Evaluation is based on the **quality of review** that you write, not on the feedback that your project receives.
- The review should be **max 250 words** answering the following questions:
 - Briefly describe the topic, research questions and the employed methods.
 - State and briefly explain two positive comments about the work.
 - State and briefly explain two improvement suggestions.

Important

In your peer review, focus on **content**, not on the formatting or quality of the slides, for instance.

Schedule

March

13th: Getting started, Introduction to Data Science (DS)

20th: The essentials of R programming, The DS workflow – Part I: Import, Tidy and Transform

April

10th/17th: The DS workflow – Part II: Visualize

24th: The DS workflow – Part III: Model

May

8th/18th/22nd/29th: The DS workflow – Part III: Model

June

1st: The DS workflow – Part IV: Communicate

12th: Final presentations of the group projects

Deadlines

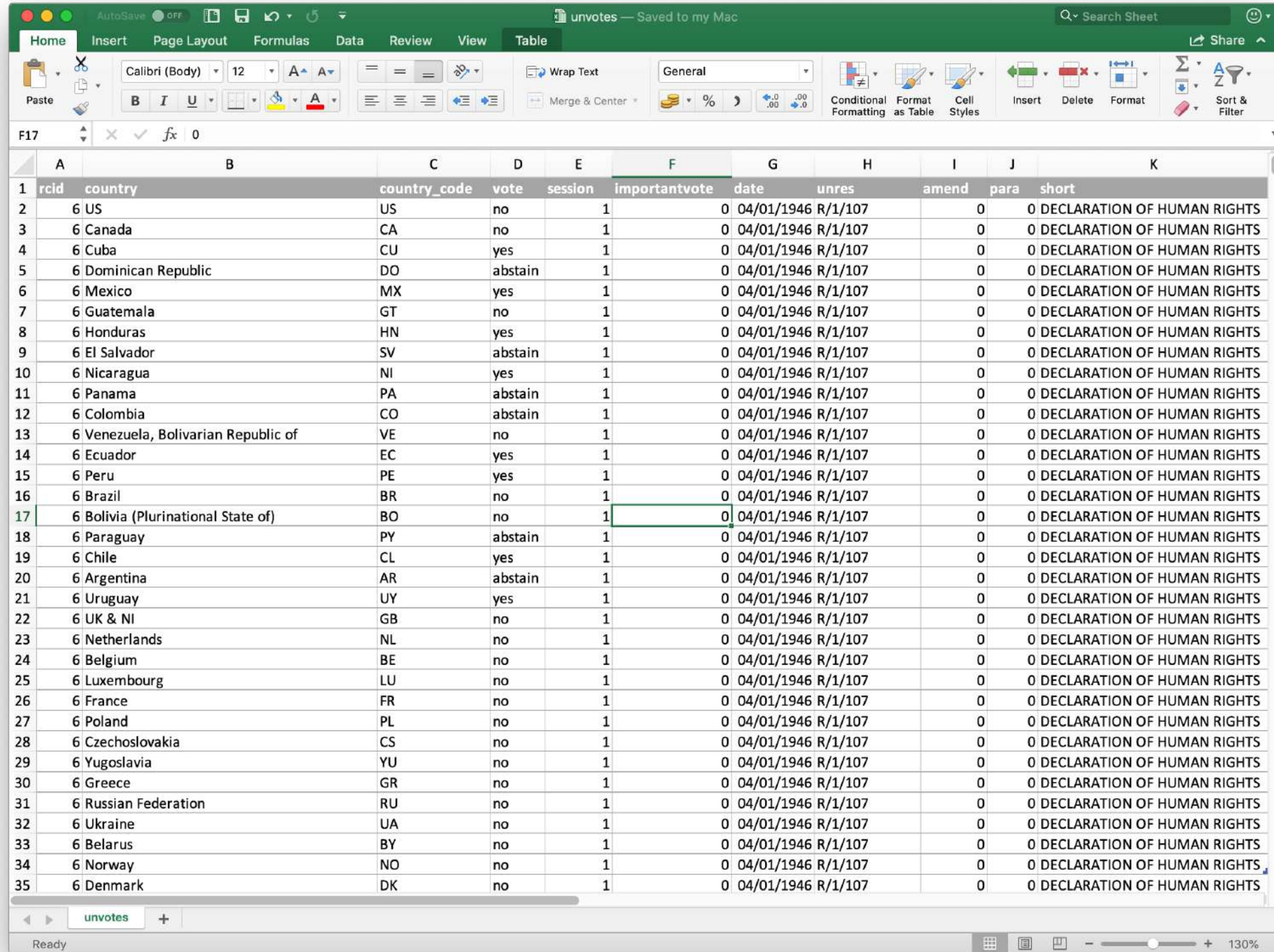
- **20th March:** organize in teams, send team members via e-mail
- **11th June:** Send presentations via e-mail
- **26th June:** Hand in group project reports and peer reviews

Questions and contact

- Any questions?
- Contact:
 - Anytime via e-mail: julianamonphd@gmail.com

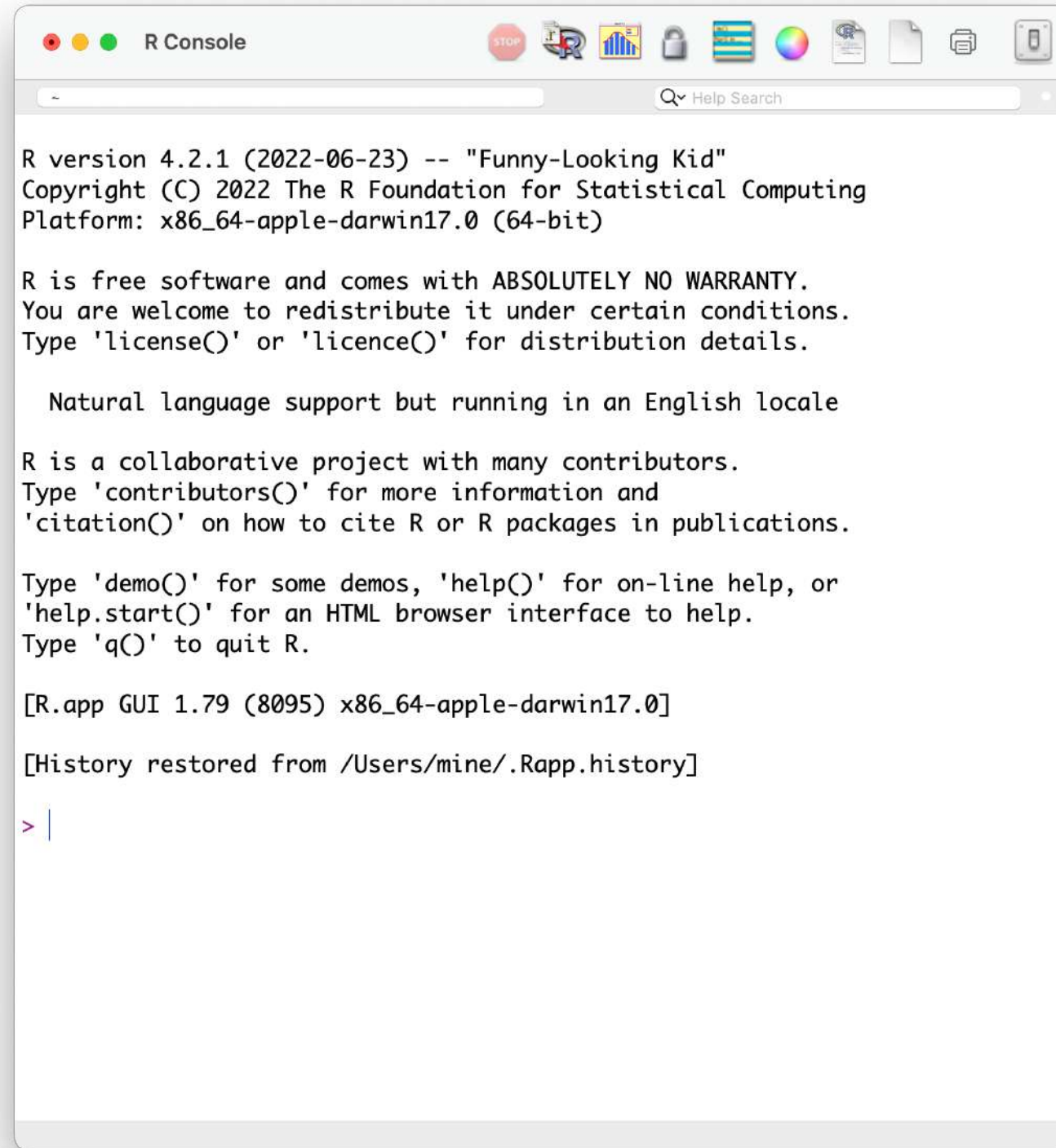
Course toolkit

Software – Excel?



	A	B	C	D	E	F	G	H	I	J	K
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2	6	Canada	CA	no	1	0	04/01/1946	R/1/107		0	DECLARATION OF HUMAN RIGHTS
3	6	Cuba	CU	yes	1	0	04/01/1946	R/1/107		0	DECLARATION OF HUMAN RIGHTS
4	6	Dominican Republic	DO	abstain	1	0	04/01/1946	R/1/107		0	DECLARATION OF HUMAN RIGHTS
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18	6	Chile	CL	yes	1	0	04/01/1946	R/1/107		0	DECLARATION OF HUMAN RIGHTS
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31	6	Ukraine	UA	no	1	0	04/01/1946	R/1/107		0	DECLARATION OF HUMAN RIGHTS
32	6	Belarus	BY	no	1	0	04/01/1946	R/1/107		0	DECLARATION OF HUMAN RIGHTS
33	6	Norway	NO	no	1	0	04/01/1946	R/1/107		0	DECLARATION OF HUMAN RIGHTS
34	6	Denmark	DK	no	1	0	04/01/1946	R/1/107		0	DECLARATION OF HUMAN RIGHTS

Software – R



The screenshot shows the R Console window on a macOS system. The title bar reads "R Console". The window contains the following text:

```
R version 4.2.1 (2022-06-23) -- "Funny-Looking Kid"
Copyright (C) 2022 The R Foundation for Statistical Computing
Platform: x86_64-apple-darwin17.0 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

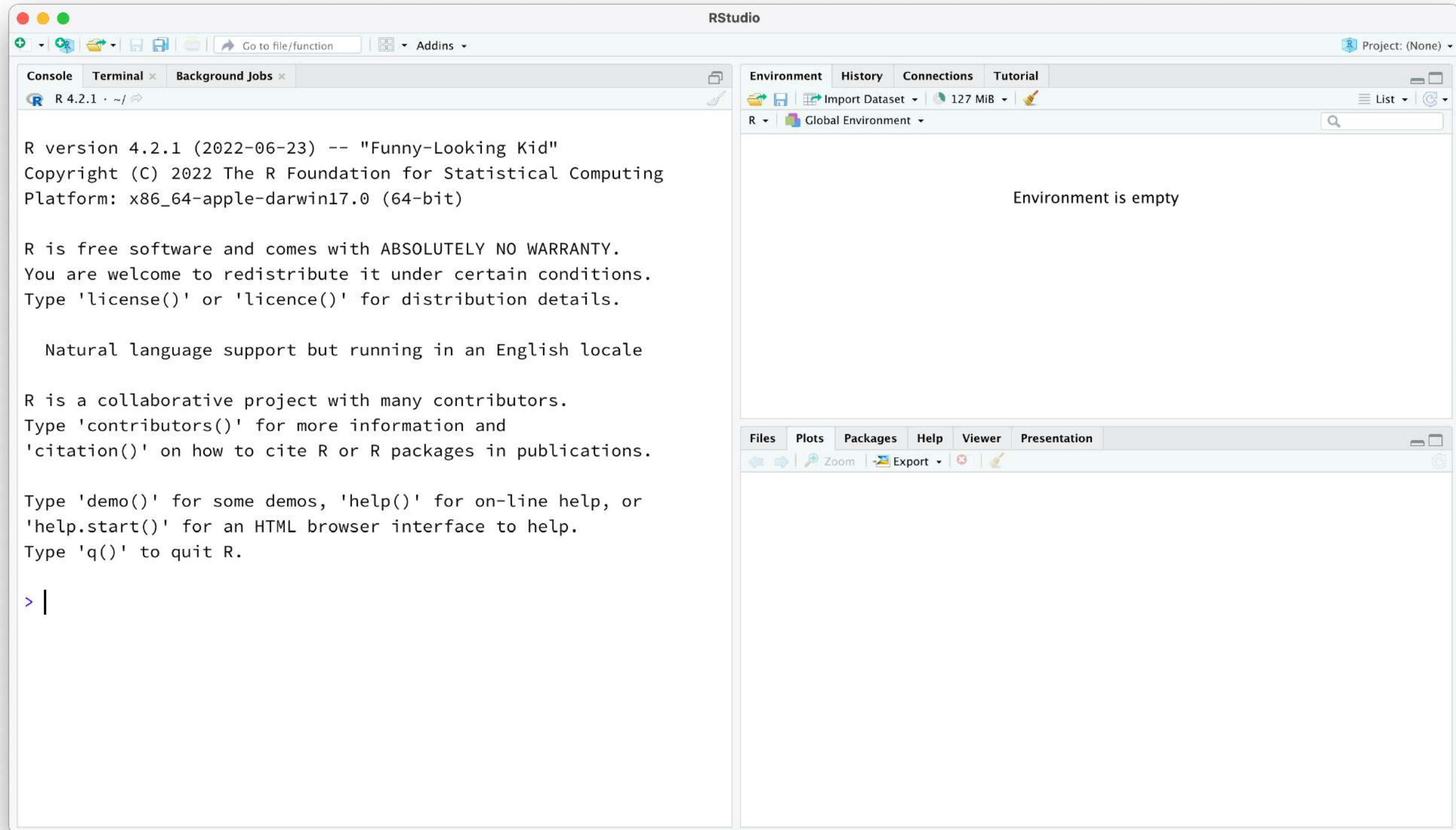
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[R.app GUI 1.79 (8095) x86_64-apple-darwin17.0]

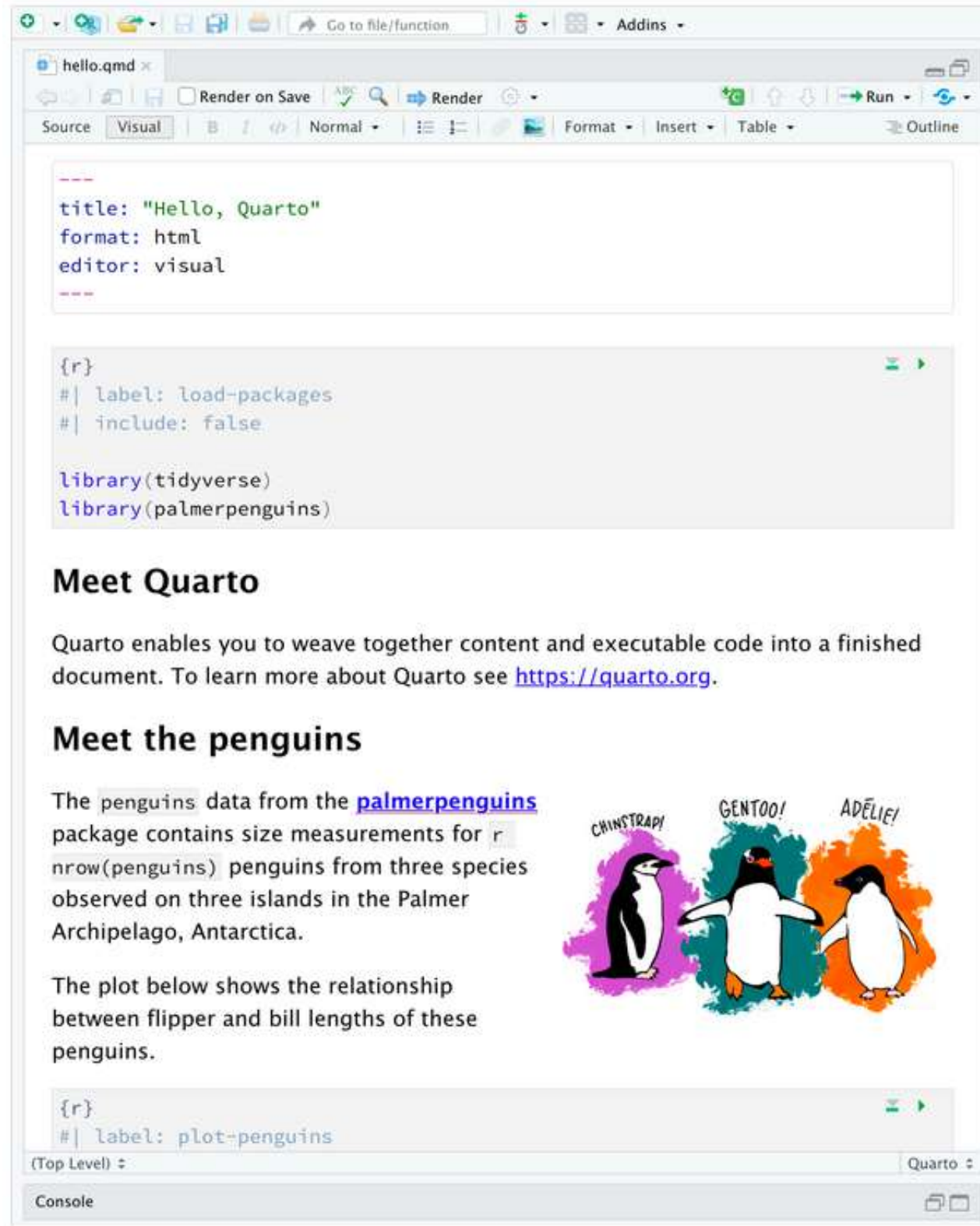
[History restored from /Users/mine/.Rapp.history]

> |
```


Software – RStudio



Software – Quarto



The screenshot shows the Quarto editor interface. The top bar includes a search bar, a file explorer, and a toolbar with icons for rendering, running, and saving. The main editor area displays the source code for a document titled "Hello, Quarto". The code is written in R and includes a title, format, editor, and a code block for loading packages and plotting penguins.

```

---
title: "Hello, Quarto"
format: html
editor: visual
---

{r}
#| label: load-packages
#| include: false

library(tidyverse)
library(palmerpenguins)


```

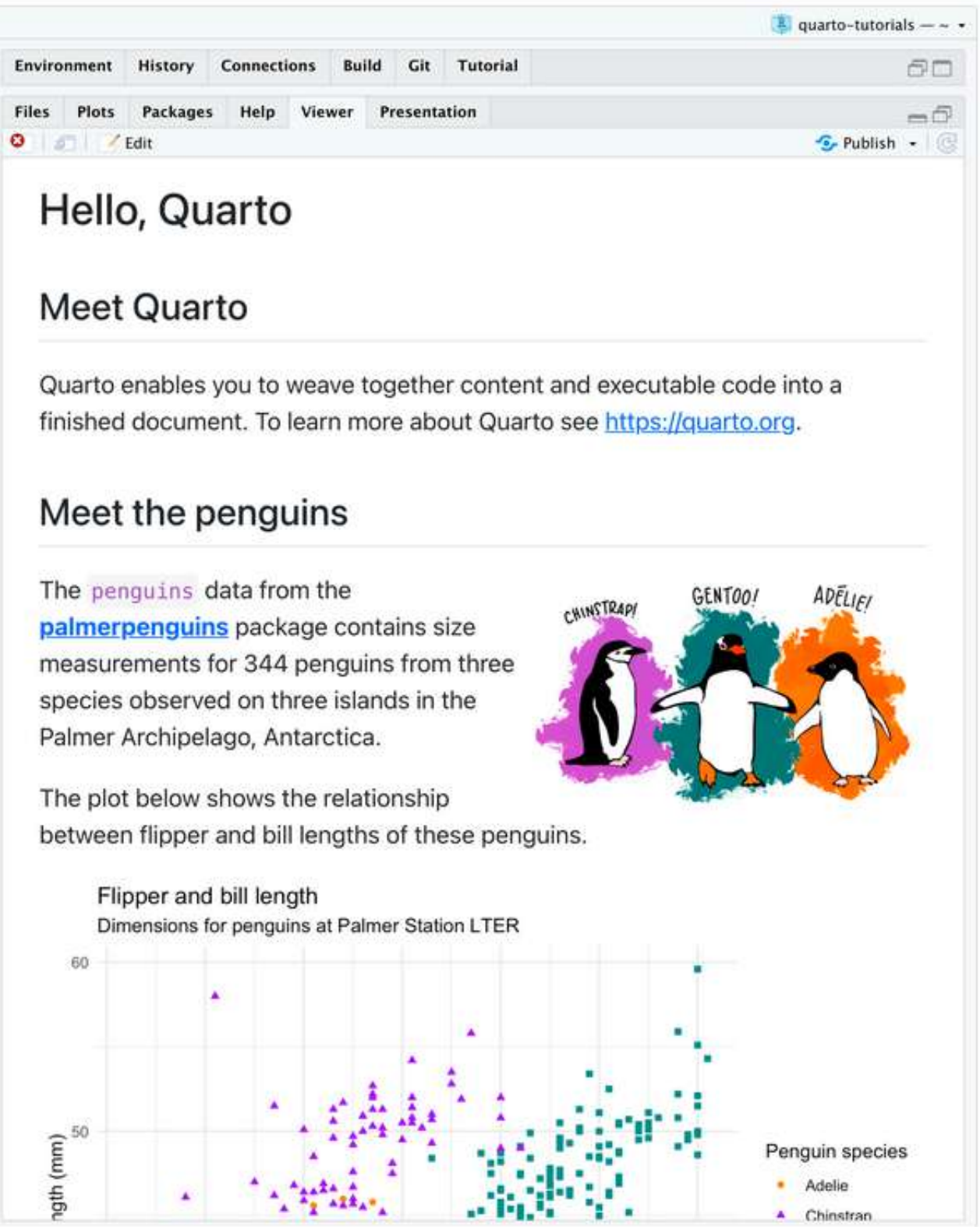
Below the code block, the rendered document is shown. It features a title "Hello, Quarto", a subtitle "Meet Quarto", and a paragraph explaining that Quarto enables you to weave together content and executable code into a finished document. To learn more about Quarto, it links to <https://quarto.org>. The next section is titled "Meet the penguins" and describes the penguins data from the palmerpenguins package, which contains size measurements for 344 penguins from three species observed on three islands in the Palmer Archipelago, Antarctica. A small illustration of three penguins (Chinstrap, Gentoo, and Adélie) is shown. The text states that the plot below shows the relationship between flipper and bill lengths of these penguins.

```

{r}
#| label: plot-penguins


```

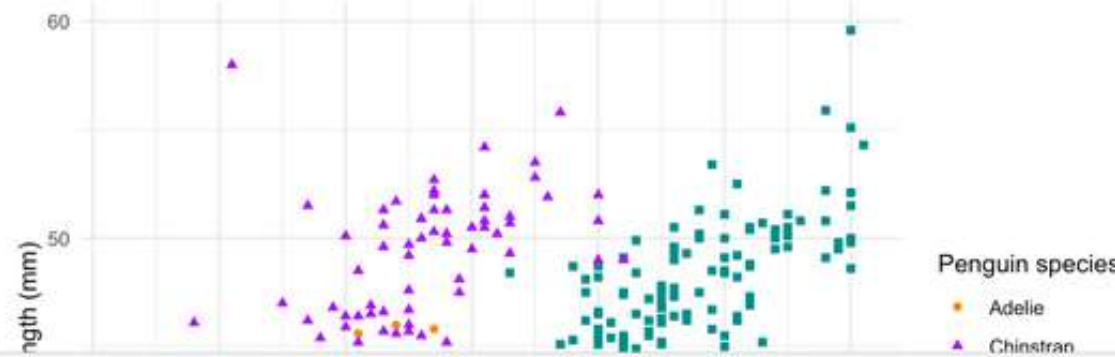
The bottom of the editor shows a console window with the text "(Top Level) :".



The screenshot shows the rendered document. It features a title "Hello, Quarto", a subtitle "Meet Quarto", and a paragraph explaining that Quarto enables you to weave together content and executable code into a finished document. To learn more about Quarto, it links to <https://quarto.org>. The next section is titled "Meet the penguins" and describes the penguins data from the palmerpenguins package, which contains size measurements for 344 penguins from three species observed on three islands in the Palmer Archipelago, Antarctica. A small illustration of three penguins (Chinstrap, Gentoo, and Adélie) is shown. The text states that the plot below shows the relationship between flipper and bill lengths of these penguins.

The plot below shows the relationship between flipper and bill lengths of these penguins.

Flipper and bill length
Dimensions for penguins at Palmer Station LTER



Penguin species

- Adélie
- Chinstrap
- Gentoo

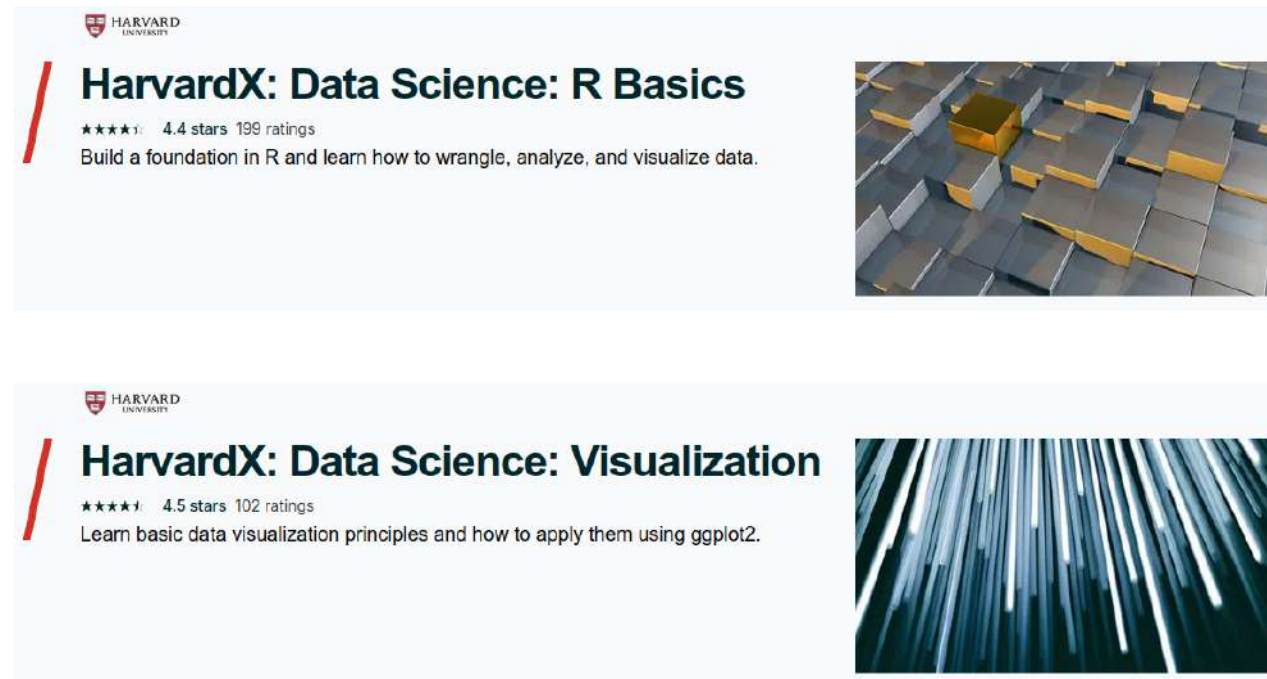
Software

- Modern data science is unthinkable without **computer programming**: typically, either [Python](#) or [R](#) is used.
- For the purposes of this course, we will use:
 - The open-source statistical programming language [R](#).
 - A bespoke integrated development environment (IDE) for R called [RStudio](#).
 - An authoring framework for creating beautiful reports, presentations, web sites, etc., combining text, code, results and visualizations, called [Quarto](#).
- Until next time, therefore please
 - either **install** R, RStudio and Quarto on your laptop (recommended) or
 - register for a free account at [Posit Cloud](#).

Resources

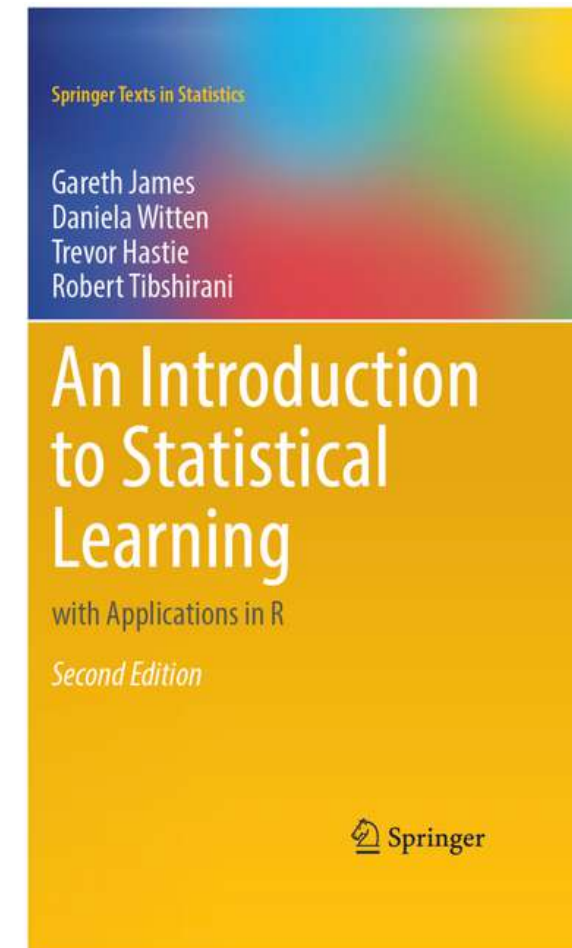
- **Primarily:** slides and exercises provided
- However, for a deeper dive and additional materials, I recommend:

For R programming



Excellent courses from Harvard Professor Rafael Irizarry, available for free [here](#) and [here](#).

For machine learning



Excellent book, available for free [here](#).

Resources – How about AI?

- With the large-scale adoption of AI tools like **ChatGPT**, the way data scientists work is rapidly changing.
- This course therefore **actively encourages** the use of AI tools for R programming. Here are some guidelines:
 - Use ChatGPT for **programming**, not for writing the project report.
 - Do **not** just copy-paste code generated by ChatGPT. Run it line-by-line, try to understand and edit as needed.
 - Engineer your prompts until the response starts to look like code you are learning in this course.
 - If the response is not correct, ask for a correction.
- With the arrival of AI, programming is becoming ever more **accessible**, but the need for people like you who actually **understand** the code they are running, is also increasing.

Resources – How about AI?

